

Dr. Inder Pal Singh

Luxembourg-based

SUMMARY

Computer Vision · Data Science · GenAI/RAG · Applied AI Systems

Luxembourg-based AI/ML professional with a Ph.D. in Computer Science and an M.S. in Computer Vision & Machine Learning. Experienced in computer vision, deep learning, data science, GenAI/RAG workflows, explainable AI, 3D vision, space imagery, and applied machine learning systems. Combines academic research experience with several years of industry experience in data analytics, technical delivery, stakeholder communication, and team leadership.

EDUCATION

University of Luxembourg <i>Ph.D. in Computer Science</i>	Luxembourg 2020 – 2024
University of Burgundy <i>M.S. in Computer Vision & Machine Learning</i>	Dijon, France 2018 – 2020
M.D. University <i>Bachelor of Engineering</i>	Rohtak, India 2005 – 2009

EXPERIENCE

Applied AI / Computer Vision / Machine Learning Researcher <i>University of Luxembourg</i>	Aug 2024 – Oct 2025 Luxembourg
- Worked on ESA-funded space imagery AI, including deep learning workflows for spacecraft 6-DoF pose estimation, synthetic-to-real adaptation, model validation, benchmarking, and failure analysis. - Developed applied ML and GenAI prototypes combining XGBoost, SHAP explainability, retrieval-augmented generation, structured AI outputs, and Streamlit-based user interfaces. - Built reproducible experimentation workflows using Python, PyTorch, TensorFlow/Keras, Scikit-learn, Docker, Git, Azure exposure, and HPC/GPU systems. - Collaborated with research, engineering, and project stakeholders to translate AI methods into usable prototypes, technical reports, and research outputs. - Supported student/project mentoring, research communication, proposal activities, and scientific documentation.	
PhD Researcher – Computer Vision & Deep Learning <i>University of Luxembourg</i>	Oct 2020 – Jul 2024 Luxembourg
- Conducted Ph.D. research on compact and effective deep learning methods for multi-label image classification, visual recognition, and domain adaptation. - Designed CNN/GCN-based models for modeling relationships between multiple visual concepts in images. - Developed and evaluated domain adaptation methods to improve robustness across changing visual domains and reduce annotation dependency. - Worked on computer vision applications including deepfake manipulation localization, fashion attribute recognition, remote sensing/space imagery, and synthetic 3D human pose data. - Performed dataset preparation, model training, validation, benchmarking, ablation studies, failure analysis, and reproducible GPU/HPC experimentation. - Published first-author research in venues including WACV, ICIP, MMSP, CVIU, ESWA, and ASTRA.	
Data Analytics & Technical Delivery Lead <i>Tata Consultancy Services</i>	Apr 2013 – Sep 2018 Sweden / India

- Led a 14-member outsourced technical delivery team onsite at TeliaSonera AB, Stockholm, ensuring SLA/KPI compliance and alignment with client expectations.
- Delivered KPI analytics, dashboard reporting, SQL-based operational reporting, and steering-level updates to support data-driven governance.
- Managed project delivery from due diligence, transition, stabilization, and BAU execution to continuous improvement.
- Developed SOPs, reporting templates, documentation, and knowledge-transfer processes to reduce errors and improve team consistency.
- Collaborated with cross-functional business, technical, and operations teams to identify bottlenecks, improve workflows, and reduce manual effort by approximately 30%.
- Built strong experience in stakeholder communication, process improvement, technical documentation, operational analytics, and team leadership.

GIS / Telecom Data Engineer

Feb 2011 – Apr 2013

Techsignia Solutions Pvt. Ltd.

India / Saudi Arabia

- Worked on telecom network engineering and planning using ESRI ArcGIS and geospatial data workflows for OSP/ISP telecom inventory.
- Supported data migration, validation, and infrastructure data modeling using Telcordia Network Engineer for Saudi Aramco.
- Used Telcordia Analyst for decision analysis, work-order approval, reporting, and structured data sharing.
- Gained experience in geospatial analytics, telecom infrastructure data, data quality checks, migration workflows, technical reporting, and client-facing delivery.

Telecom Performance Coordinator / Project Engineer

Aug 2009 – Sep 2010

Infinite Telesolutions Pvt. Ltd.

India

- Performed drive-test surveys, site verification, and physical optimization activities for telecom network sites.
- Collected and analyzed field performance data to identify coverage, handover, quality, and drop-call related problem areas.
- Prepared technical reports and coordinated resources, meetings, field teams, and project updates.
- Built early experience in telecom operations, field data analysis, network optimization, project coordination, and client reporting.

PROJECTS

ESA Space AI & 6-DoF Pose Estimation | Deep learning workflows for spacecraft pose estimation, synthetic-to-real adaptation, model validation, benchmarking, and robustness analysis.

AML AI Reporter / GenAI Decision-Support Prototype | Applied ML and GenAI system combining XGBoost, SHAP, LangChain/RAG, vector retrieval, structured reports, citations, Streamlit UI, Dockerized workflows, and explainable recommendations.

Multi-Label Image Classification & Domain Adaptation | Ph.D. research on CNN/GCN-based visual recognition, graph-based label relationships, domain adaptation, and robust multi-label classification.

Synthetic 3D Human Pose Dataset | Generated synthetic human pose data using 3D body scans, virtual backgrounds, and arbitrary camera viewpoints to improve 2D/3D pose estimation robustness.

Deepfake Manipulation Localization | Worked on multi-label deepfake classification and localization, focusing on fine-grained manipulation cues and explainable visual evidence.

Telecom KPI Analytics & Process Optimization | Delivered KPI reporting, operational analytics, SLA governance, stakeholder dashboards, and workflow improvements in large-scale telecom delivery environments.

TECHNICAL SKILLS

Machine Learning & Data Science: Regression, classification, predictive analytics, feature engineering, model validation, benchmarking, explainability, XGBoost, SHAP, Scikit-learn.

Computer Vision: Image classification, multi-label classification, object detection exposure, pose estimation, 6-DoF pose estimation, deepfake localization, domain adaptation, synthetic data, OpenCV.

3D Vision & Robotics Exposure: Open3D, point clouds, 3D scanner data, synthetic 3D pose data, ROS exposure, robotics-oriented perception workflows.

GenAI / LLM / RAG: LangChain, OpenAI API, Chroma, semantic retrieval, prompt engineering, structured outputs, RAG-based reporting, LLM-powered decision support.

AI Engineering & Deployment: Streamlit applications, Dockerized workflows, Git/GitHub, API integration awareness, Linux/SSH, reproducible experiments, cloud/HPC/GPU workflows.

Data & Reporting: SQL-based reporting, KPI analytics, dashboarding, data validation, technical documentation, stakeholder reporting, business insight communication.

Languages: Python, SQL, Bash

ML/DL: PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, SHAP

Computer Vision: OpenCV, Open3D, CNNs, GCNs, pose estimation workflows

GenAI/RAG: LangChain, OpenAI API, Chroma, prompt engineering

Deployment/Engineering: Streamlit, Docker, Git, GitHub, Linux, SSH, VS Code

Cloud/Compute: Azure exposure, AWS exposure, HPC/GPU systems

Ramp-up Areas: FastAPI, MLflow, ONNX, TensorRT, NVIDIA Jetson, CUDA, C++ production workflows

CERTIFICATIONS

Microsoft Azure AI Fundamentals — Passed 2026

Azure AI Engineer Associate — In progress

Azure ML Operations Engineer Associate — In progress

PROFESSIONAL COMPETENCIES

Applied AI & Machine Learning: Experience developing ML and deep learning workflows for predictive modeling, visual recognition, pose estimation, explainability, and decision-support applications.

Computer Vision & 3D AI: Experience in multi-label image classification, domain adaptation, 6-DoF spacecraft pose estimation, deepfake localization, synthetic 3D pose data, Open3D, and point cloud handling.

GenAI, LLMs & RAG Systems: Built applied GenAI prototypes combining retrieval, structured prompting, LLM-generated reports, vector databases, and ML-based risk prediction.

Data Science & Explainable AI: Experience with XGBoost, SHAP, model evaluation, feature engineering, benchmarking, statistical thinking, and translating model outputs into actionable insights.

Deployment-Oriented AI Prototyping: Experience with Streamlit-based AI applications, Dockerized workflows, Git-based version control, API-based integration awareness, Linux/SSH workflows, cloud exposure, and GPU/HPC experimentation.

Research, Communication & Delivery: First-author publications, scientific writing, teaching, mentoring, technical documentation, stakeholder reporting, client-facing delivery, and cross-functional collaboration.

PUBLICATIONS

First-author and co-author publications in computer vision, multi-label image classification, domain adaptation, deepfake detection, and space AI. Venues include WACV, ICIP, MMSP, CVIU, ESWA, AAAI Workshop, and ASTRA.

LANGUAGES

English: Fluent French: Intermediate Luxembourgish: Basic / learning Hindi: Native